Mathematical Formulations and Governing Equations for WAAM Numerical Investigation

Concise Formulation Compendium: Thermal, Mechanical, Source, and Discretization Models

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1. Thermal Energy Transport and Phase Change Formulations

Non-Linear Transient Heat Conduction Equation (Strong Form)

$$\rho(T)\, c_p(T)\, \frac{\partial T}{\partial t} = \nabla \cdot \big(k(T)\, \nabla T\big) + q_{\text{source}}(\vec{x},t) - \rho(T)\, L_f\, \frac{\partial f_l}{\partial t}$$

Mushy Zone Liquid Fraction Linear Evolution

$$f_l(T) = \begin{cases} 0, & T < T_{\text{sol}} \\ \dfrac{T - T_{\text{sol}}}{T_{\text{liq}} - T_{\text{sol}}}, & T_{\text{sol}} \le T \le T_{\text{liq}} \\ 1, & T > T_{\text{liq}} \end{cases}$$

Apparent Heat Capacity Formulation

$$c_p^*(T) = c_p(T) + L_f\, \frac{\partial f_l}{\partial T} = \begin{cases} c_p(T), & T < T_{\text{sol}} \text{ or } T > T_{\text{liq}} \\ c_p(T) + \dfrac{L_f}{T_{\text{liq}} - T_{\text{sol}}}, & T_{\text{sol}} \le T \le T_{\text{liq}} \end{cases}$$

Enthalpy Formulation with Apparent Heat Capacity

$$\rho(T)\, c_p^*(T)\, \frac{\partial T}{\partial t} = \nabla \cdot \big(k(T)\, \nabla T\big) + q_{\text{source}}(\vec{x},t)$$

Finite Element Variational Weak Form (Backward Euler, $\theta = 1.0$)

$$\int_\Omega \left(\frac{\rho\, c_p^*}{\Delta t}\, T^{n+1}\, v + k(T)\, \nabla T^{n+1} \cdot \nabla v \right) d\Omega + \int_{\partial\Omega} h_{\text{eff}}\, T^{n+1}\, v\, dA = \int_\Omega \left(\frac{\rho\, c_p^*}{\Delta t}\, T^n + q_{\text{source}}^{n+1} \right) v\, d\Omega + \int_{\partial\Omega} h_{\text{eff}}\, T_\infty\, v\, dA$$

Linearized Stefan-Boltzmann Radiation Coefficient

$$h_{\text{rad}}(T) = \varepsilon\, \sigma_{\text{SB}}\, (T + T_\infty)\, (T^2 + T_\infty^2) \approx 4\, \varepsilon\, \sigma_{\text{SB}}\, T_{\text{ref}}^3$$

Effective Combined Surface Convective-Radiative Heat Transfer Coefficient

$$h_{\text{eff}} = h_{\text{conv}} + h_{\text{rad}}$$

Boiling Temperature Surface Evaporation Cap

$$T(\vec{x},t) = \min\big(T(\vec{x},t),\, T_{\text{boil}}\big) \qquad (T_{\text{boil}} = 2862\,^\circ\text{C})$$

Latent Heat of Vaporization Dissipation Accounting

$$E_{\text{cap}} = \int_\Omega \rho\, c_p^*\, \max(T - T_{\text{boil}},\, 0)\, d\Omega$$

2. Moving Volumetric Heat Source Formulations (Goldak Double-Ellipsoid)

Lagrangian-to-Eulerian Torch Coordinate Transformation

$$\xi = (x - x_0) - v_{\text{scan}}\,(t - t_{\text{start}})$$

Front Semi-Ellipsoid Heat Flux Density Distribution ($\xi \geq 0$)

$$q_f(\xi, y, z) = \frac{6\sqrt{3}\, f_f\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_f\, b_G\, c_G} \exp\left(-3\,\frac{\xi^2}{a_f^2} - 3\,\frac{y^2}{b_G^2} - 3\,\frac{(z - z_0)^2}{c_G^2}\right)$$

Rear Semi-Ellipsoid Heat Flux Density Distribution ($\xi < 0$)

$$q_r(\xi, y, z) = \frac{6\sqrt{3}\, f_r\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_r\, b_G\, c_G} \exp\left(-3\,\frac{\xi^2}{a_r^2} - 3\,\frac{y^2}{b_G^2} - 3\,\frac{(z - z_0)^2}{c_G^2}\right)$$

Heat Flux Spatial Continuity Constraint at Arc Center Plane ($\xi = 0$)

$$q_f(0, y, z) = q_r(0, y, z) \iff \frac{f_f}{a_f} = \frac{f_r}{a_r} \implies f_r = f_f\left(\frac{a_r}{a_f}\right)$$

Total Power Energy Conservation Constraint

$$\int_{-\infty}^{0}\iint_{-\infty}^{\infty} q_r\, dy\, dz\, d\xi + \int_{0}^{\infty}\iint_{-\infty}^{\infty} q_f\, dy\, dz\, d\xi = Q_{\text{net}} \iff f_f + f_r = 2.0$$

Energy Partition Fractions Solution

$$f_f = \frac{2\,a_f}{a_f + a_r}, \qquad f_r = \frac{2\,a_r}{a_f + a_r}$$

Net Absorbed Arc Thermal Power

$$Q_{\text{net}} = \eta_{\text{arc}}\, U\, I$$

Transverse Semi-Axis Sizing from Bead Width

$$b_G = 0.50\, w_{\text{bead}}$$

Penetration Semi-Axis Sizing from Bead Height

$$c_G = 1.25\, h_{\text{bead}}$$

Longitudinal Front Semi-Axis Sizing

$$a_f = 0.50\, w_{\text{bead}}$$

Longitudinal Rear Semi-Axis Sizing

$$a_r = 1.25\, w_{\text{bead}} = 2.50\, a_f$$

Dimensionless Energy Fractions for 2.5x Rear Elongation

$$f_f = \frac{2(0.50\, w_{\text{bead}})}{0.50\, w_{\text{bead}} + 1.25\, w_{\text{bead}}} = \frac{1.00}{1.75} \approx 0.60, \qquad f_r = \frac{2(1.25\, w_{\text{bead}})}{0.50\, w_{\text{bead}} + 1.25\, w_{\text{bead}}} = \frac{2.50}{1.75} \approx 1.40$$

Peak Volumetric Power Density at Torch Axis ($\xi = 0, y = 0, z = z_0$)

$$q_0 = \frac{6\sqrt{3}\, f_f\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_f\, b_G\, c_G} = \frac{6\sqrt{3}\, f_r\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_r\, b_G\, c_G}$$

3. Quiet Element Activation and Inactive State Suppression

Element Activation State Heaviside Function

$$\psi(\vec{x},t) = \begin{cases} 1, & \vec{x} \in \Omega_{\text{activated}}(t) \\ 0, & \vec{x} \in \Omega_{\text{quiet}}(t) \end{cases}$$

Progressive Kinematic Torch Activation Condition

$$\psi_e(t) = 1 \quad \Longleftrightarrow \quad x_{\text{torch}}(t) \geq x_e \quad \text{and} \quad z_{\text{torch}}(t) \geq z_e$$

Quiet Element Thermal Conductivity Degradation

$$k_{\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,k(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-4}\,k(T)$$

Volumetric Heat Capacity Conservation in Quiet Elements

$$(\rho\, c_p^*)_{\text{eff}}(\vec{x},T) = \rho(T)\, c_p^*(T)$$

Quiet Element Mechanical Stiffness Degradation

$$E_{\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,E(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-4}\,E(T)$$

Quiet Element Yield Stress Suppression

$$\sigma_{y,\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,\sigma_y(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-3}\,\sigma_y(T)$$

4. Thermal Boundary Energy Balance Equations

Top Active Surface Energy Balance (Influx, Convection, and Radiation)

$$-k(T)\,\nabla T\cdot\vec{n} = q_{\text{source}}(\xi,y,z) - h_{\text{conv}}\,(T-T_{\infty}) - \varepsilon\,\sigma_{\text{SB}}\,(T^4-T_{\infty}^4)$$

Lateral Wall Faces Convective and Radiative Dissipation (Robin BC)

$$-k(T)\,\frac{\partial T}{\partial n} = h_{\text{conv}}\,(T-T_{\infty}) + \varepsilon\,\sigma_{\text{SB}}\,(T^4-T_{\infty}^4)$$

Multi-Bead Overlapping Lateral Solid Conduction Continuity

$$-k_1\,\nabla T_1\cdot\vec{n}_1 = -k_2\,\nabla T_2\cdot\vec{n}_2, \qquad T_1 = T_2$$

Lateral Chill Block / Copper Shoe Contact Heat Flux

$$-k(T)\,\nabla T\cdot\vec{n} = h_{\text{chill}}\,(T-T_{\text{chill}})$$

Substrate Bottom Contact Heat Conduction with Fixture Table

$$-k(T)\,\nabla T\cdot\vec{n} = h_{\text{contact}}\,(T-T_{\text{table}})$$

Initial Domain Temperature Field

$$T(\vec{x},0) = T_{\text{amb}} = 25\,^{\circ}\text{C}$$

5. Feedstock Kinematics, Mass Conservation, and Bead Dimensions

Volumetric Wire Feed Delivery Rate

$$Q_{\text{wire}} = \left(\frac{\pi d_{\text{wire}}^2}{4} \right) v_{\text{wire}}$$

Theoretical Bead Cross-Sectional Area (Volume Conservation)

$$A_{\text{bead}} = \frac{Q_{\text{wire}}}{v_{\text{scan}}} = \frac{\pi d_{\text{wire}}^2 v_{\text{wire}}}{4 v_{\text{scan}}}$$

Parabolic Bead Profile Area Relation

$$A_{\text{bead}} = \frac{2}{3} w_{\text{bead}} h_{\text{bead}}$$

Parabolic Bead Reinforcement Height

$$h_{\text{bead}} = \frac{3 A_{\text{bead}}}{2 w_{\text{bead}}} = \frac{3 \pi d_{\text{wire}}^2 v_{\text{wire}}}{8 w_{\text{bead}} v_{\text{scan}}}$$

Semi-Elliptical Bead Reinforcement Height

$$h_{\text{bead}} = \frac{4 A_{\text{bead}}}{\pi w_{\text{bead}}} = \frac{d_{\text{wire}}^2 v_{\text{wire}}}{w_{\text{bead}} v_{\text{scan}}}$$

Subsequent Pass Layer Step Height (Remelting Overlap)

$$h_{\text{subseq}} = \eta_{\text{remelt}} h_{\text{bead}} \qquad (\eta_{\text{remelt}} \approx 0.80)$$

Total Multi-Layer Wall Height Accumulation

$$H_{\text{total}} = h_{\text{bead_first}} + (N_{\text{layers}} - 1) h_{\text{subseq}}$$

6. Linear Heat Input and Analytic Cooling Times

Net Linear Heat Input per Unit Travel

$$H_{\text{net}} = \frac{Q_{\text{net}}}{v_{\text{scan}}} = \frac{\eta_{\text{arc}}\, U\, I}{v_{\text{scan}}}$$

2D Thin-Plate Cooling Time ($\Delta t_{8/5}^{(2D)}$)

$$\Delta t_{8/5}^{(2D)} = \frac{(H_{\text{net}})^2}{4\,\pi\, k\, \rho\, c_p\, w_{\text{bead}}^2} \left[\left(\frac{1}{500 - T_0} \right)^2 - \left(\frac{1}{800 - T_0} \right)^2 \right]$$

3D Thick-Plate Cooling Time ($\Delta t_{8/5}^{(3D)}$)

$$\Delta t_{8/5}^{(3D)} = \frac{H_{\text{net}}}{2\,\pi\, k} \left[\frac{1}{500 - T_0} - \frac{1}{800 - T_0} \right]$$

Relative Plate Thickness Dimensional Transition Parameter

$$\theta_t = w_{\text{bead}} \sqrt{\frac{\rho\, c_p\, (500 - T_0)}{H_{\text{net}}}}$$

7. Solidification Dynamics and Microstructure Metrics

Thermal Gradient Vector and Magnitude at Solidus Front

$$\vec{G} = \nabla T, \qquad G = \|\nabla T\| = \sqrt{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2 + \left(\frac{\partial T}{\partial z}\right)^2}$$

Solidification Front Normal Growth Velocity

$$R = v_{\text{scan}} \cos\theta = v_{\text{scan}} \left(\frac{-\partial T/\partial x}{\|\nabla T\|}\right)$$

Solidification Cooling Rate

$$\dot{T} = \left|\frac{\partial T}{\partial t}\right| = G \cdot R = -v_{\text{scan}} \frac{\partial T}{\partial x}$$

Solidification Morphology Ratio

$$\frac{G}{R} = \frac{\|\nabla T\|^2}{-v_{\text{scan}} (\partial T/\partial x)}$$

Secondary Dendrite Arm Spacing (SDAS, λ_2)

$$\lambda_2 = K \left(\dot{T}\right)^{-n}$$

Columnar-to-Equiaxed Transition (Hunt CET Criterion)

$$\Phi_{\text{eq}} = \exp\left(-\frac{4\pi}{3} N_0 \left[\frac{G}{\Delta T_{\text{crit}}}\right]^{-3}\right)$$

8. Quasi-Static Mechanics and Continuum Kinematics

Quasi-Static Momentum Balance Equation (Strong Form)

$$\nabla \cdot \boldsymbol{\sigma} + \vec{b} = \vec{0} \qquad (\vec{b} \approx \vec{0})$$

Weak Variational Form of Momentum Balance

$$\int_{\Omega} \boldsymbol{\sigma} : \nabla \vec{w} \, d\Omega = \int_{\partial\Omega} \vec{t} \cdot \vec{w} \, dA + \int_{\Omega} \vec{b} \cdot \vec{w} \, d\Omega \quad \forall \vec{w} \in \mathscr{V}_0$$

Infinitesimal Strain-Displacement Relation

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla \vec{u} + (\nabla \vec{u})^T\right), \qquad \varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Additive Strain Tensor Decomposition

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p + \boldsymbol{\varepsilon}^{\text{th}}$$

Isotropic Thermal Expansion Strain Tensor

$$\boldsymbol{\varepsilon}^{\text{th}} = \alpha_{\text{sec}}(T)\,(T - T_{\text{ref}})\,\mathbf{I}$$

Secant Thermal Expansion Coefficient Definition

$$\alpha_{\text{sec}}(T) = \frac{1}{T - T_{\text{ref}}} \int_{T_{\text{ref}}}^{T} \alpha_{\text{inst}}(T')\, dT'$$

9. Constitutive Elastoplasticity and Radial Return Mapping (J_2 Flow)

Generalized Hooke's Law for Isotropic Elasticity

$$\sigma = \mathbf{C}(T) : \varepsilon^e = 2\,G(T)\,\mathbf{e}^e + K(T)\,\mathrm{tr}(\varepsilon^e)\,\mathbf{I}$$

Elastic Shear and Bulk Moduli

$$G(T) = \frac{E(T)}{2(1+\nu)}, \qquad K(T) = \frac{E(T)}{3(1-2\nu)}$$

Deviatoric Cauchy Stress Tensor

$$\mathbf{s} = \sigma - \frac{1}{3}\,\mathrm{tr}(\sigma)\,\mathbf{I}$$

Second Invariant of Deviatoric Stress (J_2)

$$J_2 = \frac{1}{2}\,\mathbf{s} : \mathbf{s} = \frac{1}{2}\,s_{ij}\,s_{ij}$$

Von Mises Equivalent Stress

$$\sigma_{\mathrm{eq}} = \sqrt{3J_2} = \sqrt{\frac{3}{2}\,\mathbf{s} : \mathbf{s}} = \sqrt{\frac{3}{2}\,s_{ij}\,s_{ij}}$$

Von Mises Yield Function with Linear Isotropic Hardening

$$f(\sigma, p) = \sigma_{\mathrm{eq}} - \sigma_y(T, p) \leq 0$$

Current Yield Strength Evolution

$$\sigma_y(T, p) = \sigma_{y0}(T) + H'(T)\,p$$

Prandtl-Reuss Associative Plastic Flow Rule

$$\dot{\varepsilon}^p = \dot{\gamma}\,\frac{\partial f}{\partial \sigma} = \dot{\gamma}\sqrt{\frac{3}{2}}\,\frac{\mathbf{s}}{\|\mathbf{s}\|} = \frac{3}{2}\,\dot{p}\,\frac{\mathbf{s}}{\sigma_{\mathrm{eq}}}$$

Accumulated Equivalent Plastic Strain Rate

$$\dot{p} = \sqrt{\frac{2}{3}\,\dot{\varepsilon}^p : \dot{\varepsilon}^p}$$

9. Constitutive Elastoplasticity and Radial Return Mapping (Continued)

Trial Elastic Strain Tensor Increment (Step $n+1$)

$$\varepsilon_{n+1}^{e,\text{trial}} = \varepsilon_{n+1} - \varepsilon_n^p - \varepsilon_{n+1}^{\text{th}}$$

Trial Cauchy and Deviatoric Stress Tensors

$$\sigma^{\text{trial}} = \mathbf{C}(T_{n+1}) : \varepsilon_{n+1}^{e,\text{trial}}, \qquad \mathbf{s}^{\text{trial}} = \sigma^{\text{trial}} - \frac{1}{3}\,\text{tr}(\sigma^{\text{trial}})\,\mathbf{I}$$

Trial Von Mises Equivalent Stress

$$\sigma_{\text{eq}}^{\text{trial}} = \sqrt{\frac{3}{2}\,\mathbf{s}^{\text{trial}} : \mathbf{s}^{\text{trial}}}$$

Trial Yield Function Evaluation

$$f^{\text{trial}} = \sigma_{\text{eq}}^{\text{trial}} - \sigma_y(T_{n+1}, p_n)$$

Plastic Multiplier Increment (Δp)

$$\Delta p = \begin{cases} 0, & \text{if } f^{\text{trial}} \le 0 \quad \text{(Elastic Step)} \\ \dfrac{\sigma_{\text{eq}}^{\text{trial}} - \sigma_y(T_{n+1}, p_n)}{3\,G(T_{n+1}) + H'(T_{n+1})}, & \text{if } f^{\text{trial}} > 0 \quad \text{(Plastic Step)} \end{cases}$$

Deviatoric Stress Radial Return Projection

$$\mathbf{s}_{n+1} = \left(1 - \frac{3\,G(T_{n+1})\,\Delta p}{\sigma_{\text{eq}}^{\text{trial}}}\right)\mathbf{s}^{\text{trial}}$$

Updated Cauchy Stress Tensor

$$\sigma_{n+1} = \mathbf{s}_{n+1} + \frac{1}{3}\,\text{tr}(\sigma^{\text{trial}})\,\mathbf{I}$$

Updated Plastic Strain Tensor and Accumulated Plastic Strain

$$\varepsilon_{n+1}^p = \varepsilon_n^p + \frac{3}{2}\,\Delta p\,\frac{\mathbf{s}^{\text{trial}}}{\sigma_{\text{eq}}^{\text{trial}}}, \qquad p_{n+1} = p_n + \Delta p$$

Thermal Plastic Strain Annealing Reset (Liquid Fusion)

$$\text{If } T \ge T_{\text{sol}} \implies \sigma \to \mathbf{0}, \quad \varepsilon^p \to \mathbf{0}, \quad p \to 0$$

10. Mechanical Boundary Constraints

Rigid Baseplate Clamping Constraint (Dirichlet BC)

$$\vec{u}(\vec{x}) = \vec{0} \iff u_x = 0, \quad u_y = 0, \quad u_z = 0 \qquad \text{on } \Gamma_{\text{clamp}}$$

Isostatic (3-2-1) Minimal Kinematic Support Constraint

$$\vec{u}(0,0,0) = \vec{0}, \qquad u_y(L_x,0,0) = u_z(L_x,0,0) = 0, \qquad u_z(0,L_y,0) = 0$$

Unilateral Baseplate Contact Support (Signorini Condition)

$$u_z \geq 0, \qquad \sigma_{zz} \geq 0, \qquad u_z \cdot \sigma_{zz} = 0 \qquad \text{at } z = 0$$

Free Surface Traction-Free Condition (Neumann BC)

$$\sigma \cdot \vec{n} = \vec{0} \qquad \text{on } \Gamma_{\text{free}}$$

11. Finite Element Discretization and Numerical Stability Criteria

Vertical Deposit Element Sizing (Δz_{dep})

$$\Delta z_{\text{dep}} = 0.50\, h_{\text{bead}} \implies n_{z,\text{dep}} = 2 \times N_{\text{layers}}$$

Transverse Deposit Element Sizing (Δy_{dep})

$$\Delta y_{\text{dep}} \approx 0.33\, w_{\text{bead}} \implies n_{y,\text{dep}} = 3$$

Longitudinal Deposit Element Sizing (Δx_{dep})

$$\Delta x_{\text{dep}} = \frac{L_x}{n_{x,\text{dep}}} = \frac{150\,\text{mm}}{30} = 5.0\,\text{mm}$$

Thermal Courant-Friedrichs-Lewy (CFL) Advection Alignment

$$\text{CFL}_{\text{thermal}} = \frac{v_{\text{scan}}\, \Delta t_{\text{weld}}}{\Delta x_{\text{dep}}} \approx \frac{(5.0\,\text{mm/s})\,(1.0\,\text{s})}{5.0\,\text{mm}} = 1.0$$